A Formal Theory of Derivative Pricing

Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

Derivative pricing is one of the central problems of modern finance. This paper develops a formal theory of pricing derivatives from first principles. The framework integrates economics, finance, probability theory, stochastic processes, statistics, computer science, artificial intelligence, psychology, welfare economics, political economy, warfare economics, and systems theory.

We begin from the Law of One Price and arbitrage-free valuation. Classical derivative pricing emerges as a special case of a broader resource-allocation framework in which derivative contracts transfer risk, information, liquidity, and strategic optionality across time and states of nature.

The resulting theory unifies forwards, futures, options, swaps, credit derivatives, real options, strategic derivatives, and AI-assisted pricing architectures.

The paper ends with "The End"

Contents

1 Introduction

A derivative is a contingent claim whose value depends upon one or more underlying state variables.

Examples include:

- Equity options
- Commodity futures
- Interest-rate swaps
- Credit default swaps
- Currency options
- Real options
- Military procurement options
- Strategic reserve contracts

The central question is:

> What is the correct price of a derivative?

The answer lies in arbitrage, risk transfer, expectations, uncertainty, market microstructure, institutional design, and computational implementation.

2 Economic Foundations

2.1 Scarcity

All economics begins with scarcity.

Let

$$R = (R_1, R_2, \dots, R_n)$$

represent available resources.
Agents seek to maximize utility

$$U(x)$$

subject to

$$x \leq R.$$

Derivative contracts arise because future resources are uncertain.

2.2 Intertemporal Exchange

Let

$$C_t$$

denote consumption at time t.
Agents maximize

$$U = \sum_{t=0}^{T} \beta^t u(C_t).$$

Future uncertainty creates demand for contingent claims.

3 Probability and State Spaces

Definition 1. *A state space is*

$$\Omega = \{\omega_1, \omega_2, \ldots, \omega_n\}.$$

Each state has probability

$$P(\omega_i).$$

A derivative payoff is

$$X(\omega).$$

Expected value is

$$E[X] = \sum_i X(\omega_i) P(\omega_i).$$

4 The Law of One Price

Theorem 1. *If two portfolios produce identical future payoffs in every state of nature, they must have identical prices.*

Proof. Suppose

$$V_A \neq V_B.$$

Assume

$$V_A < V_B.$$

Buy portfolio A and sell portfolio B.
Initial profit equals

$$V_B - V_A > 0.$$

Future payoffs cancel exactly.

Riskless profit exists.
This is arbitrage.
Therefore

$$V_A = V_B.$$

□

5 Arbitrage-Free Pricing

Let

$$S_t$$

be the underlying asset.
Assume a derivative value

$$D_t.$$

Construct a replicating portfolio

$$\Pi_t = \Delta_t S_t + B_t.$$

If

$$\Pi_T = D_T$$

then

$$D_t = \Pi_t.$$

This principle underlies modern pricing theory.

6 Binomial Pricing Model

Assume

$$S_0 = S.$$

After one period

$$S_u = uS$$

or

$$S_d = dS.$$

Risk-neutral probability:

$$q = \frac{(1+r) - d}{u - d}.$$

Derivative value:

$$D_0 = \frac{qD_u + (1-q)D_d}{1+r}.$$

7 Continuous-Time Pricing

Assume

$$dS_t = \mu S_t dt + \sigma S_t dW_t.$$

Applying Itô calculus gives

$$dD = \frac{\partial D}{\partial t}dt + \frac{\partial D}{\partial S}dS + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 D}{\partial S^2}dt.$$

Construct a hedge

$$\Delta = \frac{\partial D}{\partial S}.$$

Arbitrage elimination yields

$$\frac{\partial D}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 D}{\partial S^2} + rS\frac{\partial D}{\partial S} - rD = 0.$$

This is the Black–Scholes PDE.

8 Black–Scholes Formula

For a European call,

$$C = SN(d_1) - Ke^{-rT}N(d_2),$$

where

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}},$$

and

$$d_2 = d_1 - \sigma\sqrt{T}.$$

9 Risk-Neutral Valuation

Theorem 2. *Under absence of arbitrage,*

$$D_0 = e^{-rT}E_Q[D_T].$$

Proof. Under the equivalent martingale measure Q,

$$e^{-rt}S_t$$

is a martingale.
Therefore

$$E_Q[e^{-rT}D_T]$$

equals the unique arbitrage-free price.

□

10 A General Theory of Derivative Pricing

We define derivative value as

$$D = f(S, t, \sigma, r, L, I, A, G, M).$$

where

S	underlying state
t	time
σ	uncertainty
r	discount rate
L	liquidity
I	information
A	ambiguity
G	geopolitical risk
M	market structure

Classical models are special cases.

11 Psychological Foundations

Behavioral finance identifies systematic biases.

Examples include

- Overconfidence
- Anchoring
- Availability bias
- Loss aversion

A behavioral pricing adjustment:

$$D_B = D + \lambda B.$$

Here

$$B$$

captures aggregate behavioral distortions.

12 Psychiatric Considerations

During crises, collective behavior may become unstable.

Market panic can be represented by

$$P_t.$$

Observed derivative prices become

$$D_t^* = D_t + \alpha P_t.$$

Periods of fear therefore produce volatility smiles, volatility clustering, and liquidity shocks.

13 Biological Foundations

Evolutionary pressures favor risk management.

A simple biological utility function:

$$U = E(W) - \frac{\gamma}{2} Var(W).$$

Derivative contracts emerge naturally as risk-reduction technologies.

14 Chemical Analogy

Consider a reaction

$$A + B \to C.$$

Similarly,

$$Risk + Contract \to Transfer.$$

Derivatives function as catalytic mechanisms that redistribute uncertainty without necessarily creating additional physical resources.

15 Physical Interpretation

Financial markets resemble thermodynamic systems.

Define financial entropy

$$H = -\sum_i p_i \ln p_i.$$

Information arrival reduces uncertainty.

Pricing continuously updates as entropy changes.

16 Political Economy

Derivative markets depend on

- Property rights
- Contract enforcement
- Central banks
- Regulatory institutions

Thus

$$D = D(X, G),$$

where G denotes governance quality.

17 Geopolitics

War and sanctions influence derivative values.

Let

$$W_t$$

represent geopolitical tension.

Then

$$D_t = D_t(W_t).$$

Examples include

- Oil futures
- Currency options
- Sovereign CDS
- Defense procurement options

18 International Relations

Cross-border derivatives connect nations through risk-sharing networks.

Let

$$A_{ij}$$

represent exposure from country i to country j.

Systemic risk becomes

$$R = f(A).$$

19 Welfare Economics

Let social welfare be

$$W = \sum_i U_i.$$

Derivatives improve welfare if

$$W_{after} > W_{before}.$$

This occurs when risk is transferred to agents more capable of bearing it.

20 Warfare Economics

Military procurement often contains embedded options.

Examples:

- Expansion clauses
- Strategic reserve contracts
- Wartime production options
- Supply guarantees

A military readiness option may be valued as

$$MRO = E_Q[\text{Strategic Payoff}].$$

21 Statistics

Volatility estimation:

$$\hat{\sigma} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(r_i - \bar{r})^2}.$$

Maximum likelihood estimation:

$$L(\theta) = \prod_i f(x_i|\theta).$$

22 Data Science

Modern derivative pricing uses

- Feature engineering
- Real-time data pipelines
- Alternative data
- High-frequency information

Prediction model:

$$\hat{D} = g(X).$$

23 Artificial Intelligence

Neural-network pricing:

$$\hat{D} = NN(X; \theta).$$

Loss function:

$$\mathcal{L} = \frac{1}{n} \sum_i (D_i - \hat{D}_i)^2.$$

23.1 Reinforcement Learning

State:

$$s_t.$$

Action:

$$a_t.$$

Reward:

$$R_t.$$

Objective:

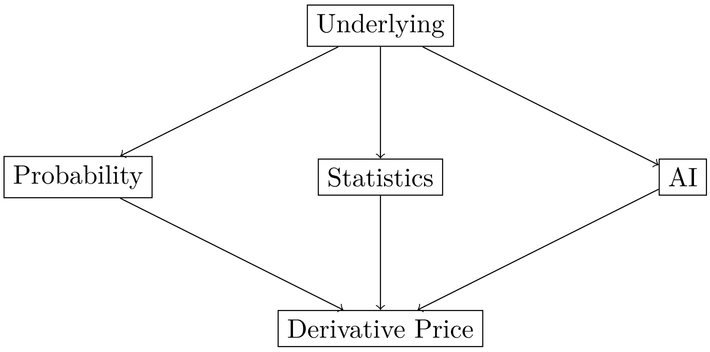
$$\max E \left[\sum_t \gamma^t R_t \right].$$

24 Algorithmic Pricing Framework

Pseudo-Code

```
Input market data

Estimate volatility

Estimate interest rate

Generate state scenarios

Compute payoff

Discount expected payoff

Return derivative value
```

25 Integrated Pricing Architecture



26 Comparative Pricing Frameworks

Method	Risk	Computation	Flexibility
Binomial	Medium	Low	Medium
Black–Scholes	Medium	Low	Low
Monte Carlo	High	High	High
Machine Learning	High	High	Very High
Reinforcement Learning	High	Very High	Very High

Table 1: Derivative Pricing Frameworks

27 Main Theorem

Theorem 3 (Formal Derivative Pricing Theorem)**.** *The value of any derivative contract equals the discounted value of its future contingent payoffs under a pricing measure adjusted for market structure, information, uncertainty, and institutional constraints.*

Formally,

$$D_0 = e^{-rT} E_Q \left[\Phi(X_T)\right] + \Psi(L, I, A, G, M).$$

Proof. Classical arbitrage-free pricing establishes

$$D_0 = e^{-rT} E_Q[\Phi(X_T)].$$

Real-world markets contain liquidity effects, information asymmetries, ambiguity, institutional constraints, and geopolitical frictions.

Represent these by

$$\Psi(L, I, A, G, M).$$

Adding these terms yields the generalized pricing equation.

Classical theory is recovered when

$$\Psi = 0.$$

□

28 Conclusion

Derivative pricing is fundamentally a problem of valuing contingent claims under uncertainty.

The classical framework emerges from arbitrage-free valuation, while the broader framework incorporates psychology, institutions, geopolitics, warfare economics, AI, and strategic decision systems.

The resulting theory provides a unified architecture for pricing derivatives across economic, financial, technological, and geopolitical domains.

References

[1] Black, F. and Scholes, M. The Pricing of Options and Corporate Liabilities. Journal of Political Economy, 1973.

[2] Merton, R. Theory of Rational Option Pricing. Bell Journal of Economics, 1973.

[3] Cox, J., Ross, S., Rubinstein, M. Option Pricing: A Simplified Approach. Journal of Financial Economics, 1979.

[4] Hull, J. Options, Futures and Other Derivatives. Pearson.

[5] Shreve, S. Stochastic Calculus for Finance. Springer.

[6] Föllmer, H. and Schied, A. Stochastic Finance. De Gruyter.

[7] Kahneman, D. Thinking, Fast and Slow.

[8] Goodfellow, I., Bengio, Y., Courville, A. Deep Learning.

[9] Sutton, R. and Barto, A. Reinforcement Learning: An Introduction.

Glossary

Arbitrage Riskless profit opportunity.

Derivative A contingent financial claim.

Martingale A process whose expected future value equals its current value.

Volatility Statistical measure of uncertainty.

Liquidity Ease of trading.

Risk-Neutral Measure Probability measure used for pricing.

Option Right but not obligation to transact.

Futures Standardized forward contract.

Swap Exchange of future cash flows.

Geopolitical Risk Risk arising from international conflict.

Strategic Optionality Value arising from future flexibility.

Machine Learning Data-driven prediction framework.

Reinforcement Learning Sequential decision optimization.

The End